Skills Summary

Writing and Using Piecewise Rules for Fees

Use this reference while completing your worksheet or when studying.

Skill 1 — Writing the rule from a price list

Shape: $C(x) = \begin{cases} \text{first formula}, & \text{first condition} \\ \text{second formula}, & \text{second condition} \end{cases}$

The second branch starts where the first stopped. If the boundary is at $x = B$ and the charge there is P, the second branch is $P + (\text{new rate})(x - B)$. Writing it as (new rate)$\times x$ throws away everything already paid — the single most common mistake on this topic.

Example: A gym charges \$30 for up to 10 visits a month, then \$2 for each extra visit. What does a month of 15 visits cost?

Step 1. The price changes at 10 visits, so this needs two branches, and 15 is past the boundary — use the second one.

Step 2. At the boundary \$30 has been paid, so the rule is $30 + 2(v - 10)$ for $v > 10$; at $v = 15$ that is $30 + 2(5)$.

$C(15) = \mathbf{40}$

Try it: A club charges \$40 for up to 12 visits, then \$3 per extra visit. Find the cost of 20 visits.

check: $C(20) = \mathbf{64}$

(!) Watch out: the conditions are part of the answer. A pair of formulas with no "when" attached does not say which one to use, and it is not a piecewise rule.

Skill 2 — Using the rule to find a charge

Two steps, in this order: test the input against the conditions to see which branch owns it, *then* substitute into that branch only.

Check a boundary value in both branches when you build a rule: a sensible price list gives the same answer either way, so the cost never jumps.

Example: A car park charges \$6 for the first hour and \$4 for each hour after that, so $C(h) = 6 + 4(h - 1)$ for $h > 1$. What does a 5-hour stay cost?
Step 1. Five hours is past the one-hour boundary, so the second branch is the one that applies — test the input before substituting.
Step 2. $C(5) = 6 + 4(5 - 1) = 6 + 16$.
$C(5) = \mathbf{22}$

Try it: Water costs \$0.15 per unit for the first 100 units, so $C(k) = 0.15k$ there. Find the cost of 80 units.

check: $C(80) = \mathbf{12}$

Skill 3 — Working back from a charge

Pick the branch first, using the boundary charge. Work out what the fee is exactly at the boundary. If the fee you were given is larger, the input lies in the upper branch; if smaller, the lower one.

Then solve that branch's equation — and check at the end that your answer really satisfies the condition you assumed.

Example: With $C(h) = 6 + 4(h - 1)$ for $h > 1$, a stay cost \$30. How many hours was it?
Step 1. At the boundary of one hour the charge is only \$6, and \$30 is larger, so the upper branch is the one to solve.
Step 2. $6 + 4(h - 1) = 30$ gives $4(h - 1) = 24$, so $h - 1 = 6$.
$h = \mathbf{7}$ hours, and $7 > 1$, so the branch was right.

Try it: With $C(n) = 20 + 2.5(n - 10)$ for $n > 10$, a bill came to \$45. Find n.

check: $n = \mathbf{20}$